

Debrup Das

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Education

- Ph.D.** *Computer Science, University of Massachusetts, Amherst, MA, USA* Sep 2024 – Ongoing
- **Advisor:** Prof. Negin Rahimi [🔗](#)
 - **Member of CIIR Lab** (Center for Intelligent Information Retrieval [🔗](#)), CIIR
Director: Prof. James Allan [🔗](#)
 - **Anticipated Graduation Date:** September 2029
- M.Sc.** *Mathematics and Computing, Indian Institute of Technology Kharagpur, India* Aug 2019 – Apr 2024
(5-year Integrated M.Sc.)
- **Thesis Advisor:** Prof. Somak Aditya [🔗](#)
 - **CGPA:** 9.00/10

Research Interests

My research focuses on building retrievers that reason about relevance rather than matching surface similarity. I am currently working on: (1) *Scaling test-time thinking for reasoning retrieval*, (2) *Reinforcement learning approaches to train retrievers optimized under embedding objectives and retrieval rewards*, (3) *Memory-augmented systems that retrieve and continually learn from past experiences*.

Publications

- Latent Personal Memory: Represent Personal Memory as Dynamic Soft Prompts** Nov 2026
Debrup Das, Avinash Amballa, Yashas Malur Saidutta, Vijay Srinivasan, Vivek Kulkarni, Srinivas Chappidi
[Findings of EMNLP 2026](#) [🔗](#)
- RaDeR: Reasoning-aware Dense Retrieval Models** Nov 2025
Debrup Das, Sam O’Nuallain, Negin Rahimi
[EMNLP 2025 Main Conference](#) [🔗](#) [Website](#) [🔗](#)
- SMAB: MAB based Word Sensitivity Estimation Framework and its Applications in Adversarial Text Generation** Apr 2025
Saurabh Kumar Pandey, Sachin Vashistha, Debrup Das, Somak Aditya, Monojit Choudhury
[NAACL 2025 Main Conference](#) [🔗](#)
- MathSensei: A Tool-Augmented Large Language Model for Mathematical Reasoning** Jun 2024
Debrup Das, Debopriyo Banerjee, Somak Aditya, Ashish Kulkarni
[NAACL 2024 Main Conference](#) [🔗](#)

Graduate Research

- Thinking Reasoning Retrievers, UMass Amherst** Research Assistant May 2026 – Current
- Research on developing strong retrievers capable of thinking, through new RL training approaches, guided by rewards specifically designed for retrieval.
 - Mentored by Prof. Negin Rahimi [🔗](#).
- Training Dense Retriever Models for Reasoning Retrieval, UMass Amherst** Research Assistant Sep 2024 – Apr 2025
- Research on developing reasoning-intensive retrievers from raw mathematical data (without any notion of document relevance), which shows strong generalizability to diverse domains.

- Mentored by [Prof. Negin Rahimi](#). Accepted as a Main Conference Paper at **EMNLP 2025**, Suzhou, China.

Industry Research Internships

Samsung Research America (SRA), AI Center – Mountain View

Research Intern
Feb 2026 – May 2026

- Research on building **latent-memory-augmented systems** for continual learning of long-context personalized user memory.
- Encoded user history as latent memory slots projected into **dynamic soft prompts** at inference, beating LoRA by 8.8% on PersonaMem v1 at **64× lower KV-cache cost**.
- Accepted at **Findings of EMNLP 2026**, Budapest. Mentored by [Srinivas Chappidi](#).

Rakuten, Language and Speech Processing Team

Research Intern
May 2023 – Dec 2023

- Research on *tool-augmented LLMs* for complex math reasoning.
- Integrating LLMs with symbolic tools such as *Python*, *Wolfram Alpha* and web search with *Bing API*.
- Accepted as a Main Conference Paper at **NAACL 2024**, Mexico City.

Academic Research Internships

Indian Institute of Technology, Kharagpur (in collaboration with MBZUAI)

Research Intern
Dec 2023 – May 2024

- Research on *sensitivity* of LLMs to different segments of the input, in classification tasks such as hate speech, sentiment analysis, etc.
- *Adversarial example generation* and *jailbreak attacks* using the notion of sensitivity. Mentored by [Prof. Monojit Choudhury](#) and [Prof. Somak Aditya](#), funded by [Microsoft AFMR](#).
- Accepted as a Main Conference Paper at **NAACL 2025**, Albuquerque, New Mexico.

McGill University, Montreal, Genome Quebec Innovation Center

Research Intern
May 2022 – Dec 2022

- Genome-wide association studies (*GWAS*) to detect rare disease-causing variants, mentored by [Prof. Simon Gravel](#).
- Funded by the *MITACS Globalink Research Internship Program*.

Indian Institute of Technology, Kharagpur

Research Intern
May 2021 – Dec 2021

- Training tree-based deep learning models (*Genetic programming*) with nature-inspired optimization methods such as evolutionary algorithms. Mentored by [Prof. Nirupam Chakrabarti](#), [Czech Technical University](#).

Languages and Technologies

Languages: Python, Bash, R, SQL, C, C++

Libraries: PyTorch, Transformers, FSDP, DeepSpeed, VeRL, NumPy, Scikit-Learn, SciPy, SymPy

Tools: Git, $\LaTeX$, HuggingFace, Wolfram Alpha

Teaching Assistant, UMass Amherst

- Fall 2026, **Information Retrieval** (CS 646) by [Prof. Negin Rahimi](#)
- Summer 2025 (Online), **Practice and Applications of Data Management** (CS 345) by [Prof. Gordon Anderson](#)
- Spring 2025, **Search Engines** (CS 446) by [Prof. Negin Rahimi](#)
- Fall 2024, **Search Engines** (CS 446) by [Prof. James Allan](#)

Service

- Reviewer: **ACM Transactions on Information Systems (TOIS)**, **NAACL 2025**.